

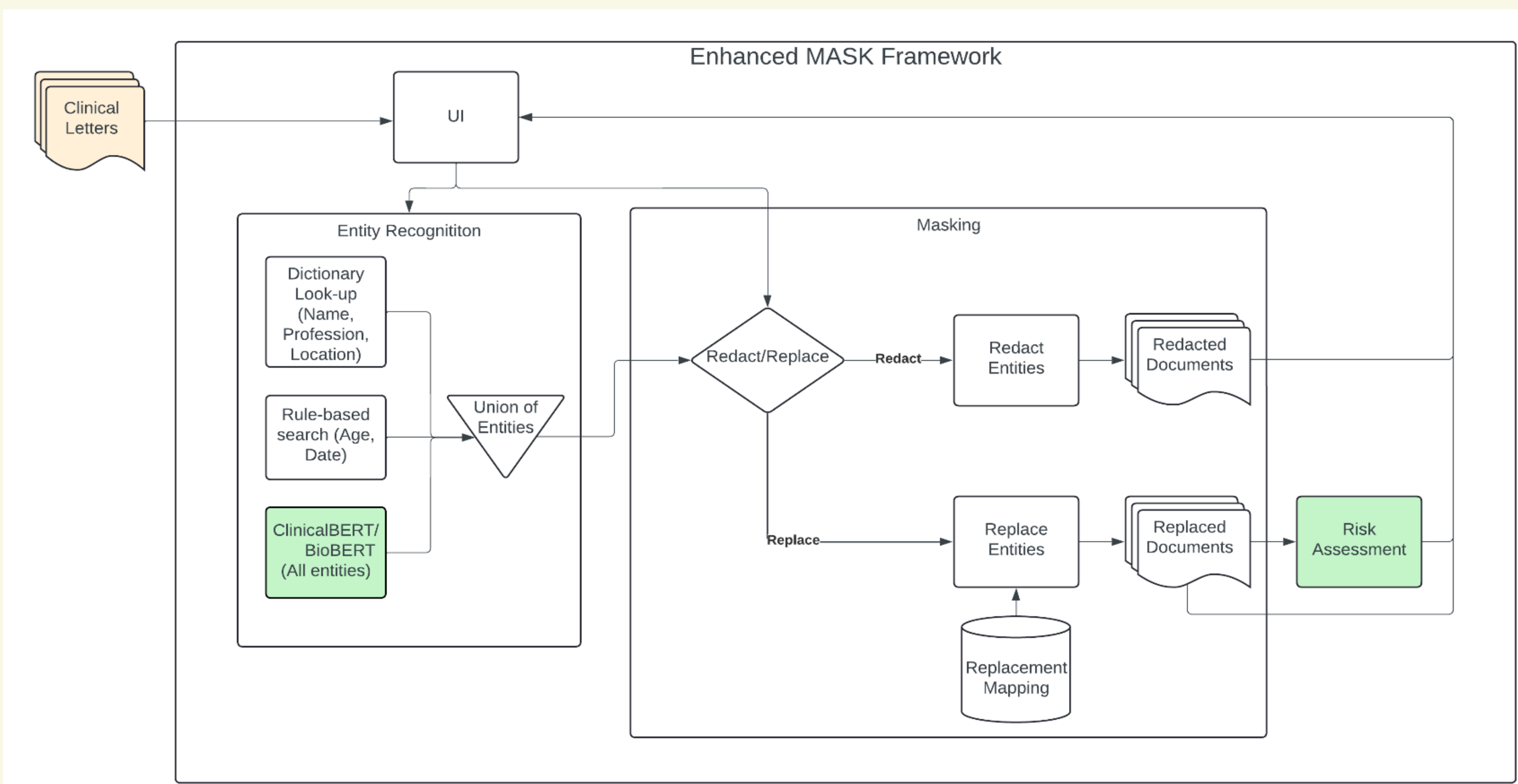
De-identifying Clinical Texts using Biomed-Clinical BERTs and Comprehensive Risk Assessment

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Motivations

- ▶ To accurately *identify* and *mask* sensitive entities (name, age, contact, IDs, location, etc.) on patients in clinical text while maintaining data utility for research.
- ▶ An advanced and user-friendly De-identification and risk assessment toolkit of clinical text by investigating state of the art language models from NLP.
- ▶ Investigating how biomedical and clinical domain-finetuned BERT models perform on this task in comparison to the traditional MASK framework.

DeID-Clinic Architecture: enhancing MASK with BioBERT and ClinicalBERT components



Methodology

- ▶ Transformer-based biomedical models **BioBERT** and **ClinicalBERT** vs traditional models: NER-BERT, NER-CRF, and LSTMs.
- ▶ Integrate BioBERT and ClinicalBERT into the **MASK** framework to facilitate effective de-identification (**MASK-BioBERT** / **ClinicalBERT**).
- ▶ **Risk assessment**: evaluate the re-identification risks of clinical data post de-identification.

Results: Different PLMs and F1-scores

Entity	BioBERT	ClinicalBERT	BiLSTM	CRF	Items
Name	98.5	97.0	97.0	96.0	622
Date	97.4	97.1	96.0	97.0	655
ID	98.8	98.9	92.0	97.0	75
Age	96.7	95.1	85.0	94.0	89
Location	92.8	92.1	91.0	90.0	278
Contact	94.1	95.4	94.0	96.0	69
Profession	74.8	73.0	79.0	63.0	27
Micro-Avg	96.5	95.9	95.0	95.0	1816
Macro-Avg	80.4	81.1	79.0	79.0	1816

Example Outputs and Interface:

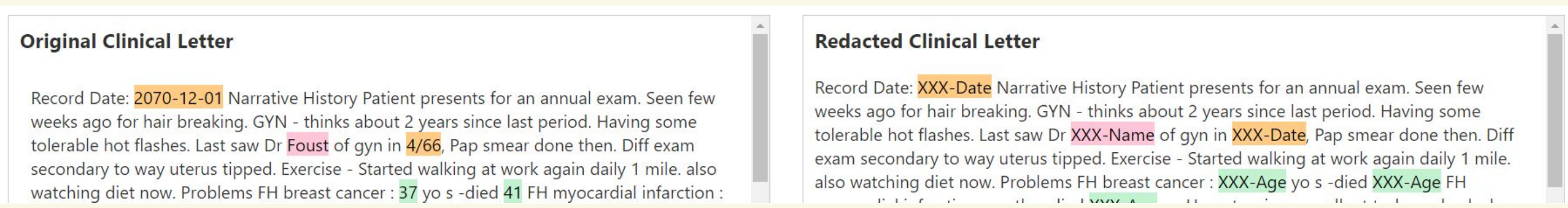


Figure: redacted clinical letter example with cancer patient data

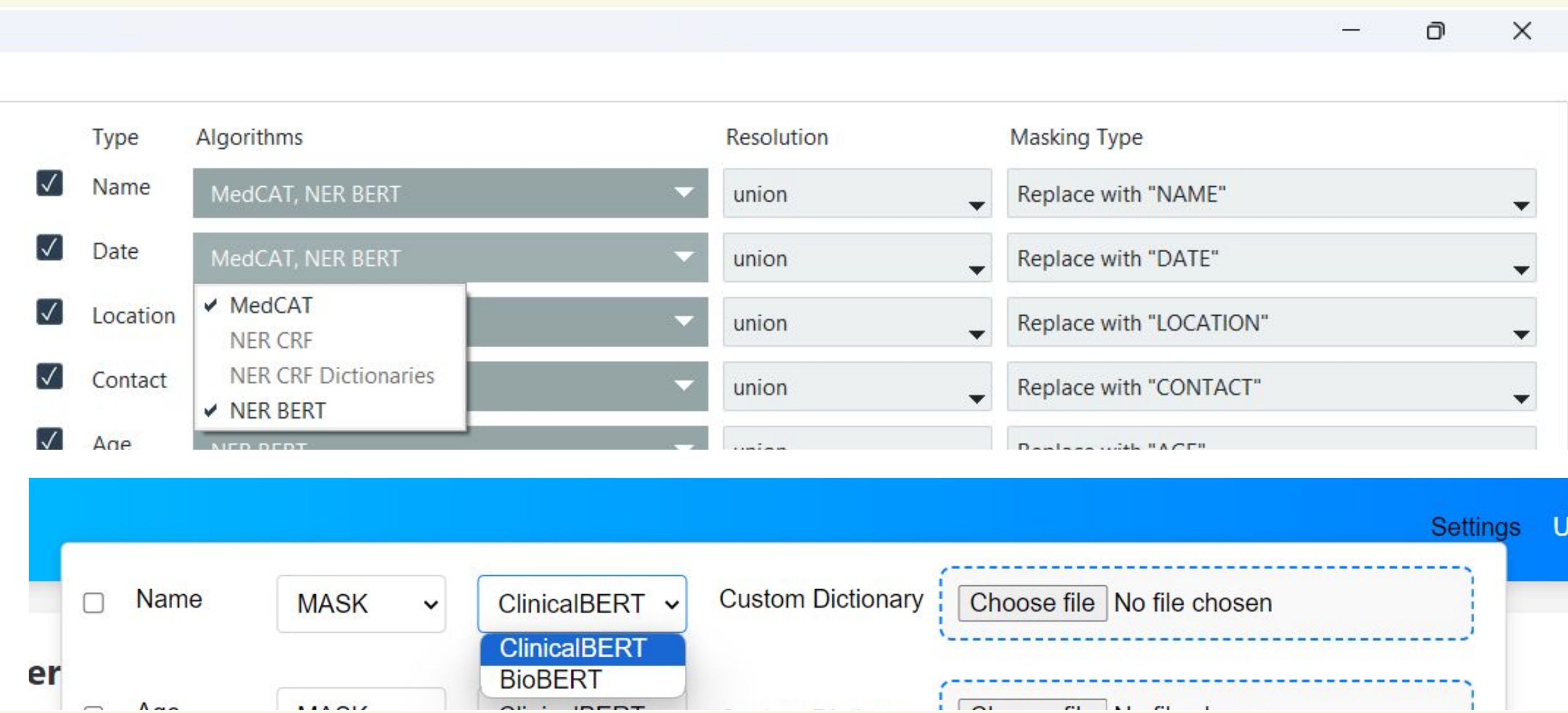


Figure: MASK framework with DeID-Clinic

Outcomes:

- ▶ BioBERT: highest Micro-average scores –vs– ClinicalBERT: highest Macro-average
- ▶ Challenging entities: **Profession** and **Contact**: future work
- ▶ A user-friendly risk-assessment platform is offered with UI design: supports different types of files.
- ▶ To carry out the deployment of our system in a real clinic setting for professional feedback.
- ▶ Our fine-tuned model and source codes: free access.

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Leiden University Medical Centre| University of Manchester | software <https://github.com/angelpaulm17/DeIDClinic>

